HH-I-1751/1
July 23, 1973

FEDERAL SPECIFICATION

INSULATION SLEEVING, THERMAL (PIPE AND TUBE COVERING) MINERAL FIBER, INDUSTRIAL TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers mineral fiber thermal insulation sleeving, (industrial type) for use on surfaces below approximately 450°F.

1.2 Classification.

1.2.1 The complete requirements for procuring Thermal Insulation Sleeving described herein shall consist of this document and the latest issue of HH-I-1751/GEN.

1.2.2 Forms, types, and classes. Mineral fiber thermal insulation sleeving shall be of the following forms, types and classes, as specified, (see 6.2).

Form A - Sleeving, pipe and tube covering.
  Type I - Molded - Three feet sections.
    Class 1 - For use at temperatures up to and including 450°F.

Form B - Molded, pipe and tube fitting covering.
  Type II - Molded, length as specified.
    Class 2 - For use at temperatures up to and including 450°F.

1.2.3 Sizes. Mineral fiber thermal insulation sleeving shall be of the sizes listed in table I, as specified (see 6.2).

TABLE I. Sizes

Form	Type	Class	Width	Length	Thickness	Thickness Increments
				Feet	Inches	
A	I	1	To fit standard	3	1/2, 3/4, 1, 1-1/2, & 2	------
B	II	2	Pipe and tube sizes as specified	As Specified	1/2, 3/4, 1, 1-1/2, & 2	------

2. APPLICABLE DOCUMENT

2.1 Latest issue of Federal Specification HH-I-1751/GEN and documents referenced herein.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 5640

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American Society for Testing and Materials (ASTM) Standards:

C 356 - Test for Linear Shrinkage of Preformed High Temperature Thermal Insulation Subjected to Soaking Heat.
C 547 - Mineral Fiber Preformed Pipe Insulation.
D 781 - Puncture and Stiffness of Paperboard, Corrugated, and Solid Fiberboard.
D 828 - Tensile Breaking Strength of Paper and Paperboard.
E 96 - Water Vapor Transmission of Materials in Sheet Form.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Forms A, and B.

3.1.1 Material. The basic material shall be fibers made from mineral substances such as rock, slag, or glass processed from a molten state, with binder, and unless otherwise specified shall meet the requirements of ASTM C 547, except for Quality Assurance, Packing, Marking and Packaging requirements.

3.1.2 Thermal conductivity. When tested in accordance with 4.4.1 of HH-I-1751/GEN thermal insulation sleeving K factors shall be in accordance with table II.

TABLE II. Thermal conductivity

Thermal conductivity Max., Btu-In/Hr. Sq. Ft. °F. At Mean Temperature	Classes 1 and 2
25	0.24
50	.25
75	.26
100	.27
150	.30
200	.40
Density, Max, Pcf	12

3.1.3 Fire hazard classification. When tested in accordance with 4.4.2 of HH-I-1751/GEN thermal insulation sleeving shall have a flame spread rating not greater than 25. The smoke developed rating shall not be greater than 50.

3.1.4 Compressive strength. When tested in accordance with 4.4.3 of HH-I-1751/GEN thermal insulation sleeving compressive strength shall withstand a straight-line squeeze at 5 percent P.S.I. deformation, or at failure, which ever is smaller.

3.1.5 Density. When tested in accordance with 4.4.4 of HH-I-1751/GEN thermal insulation sleeving density rating shall be in accordance with table II.

3.1.6 Flexural strength. When tested in accordance with 4.4.5 of HH-I-1751/GEN thermal insulation shall show no visible rupture on outer surfaces when bent, and upon release shall spring back to its original form and dimensions.

3.1.7 Temperature ratings. When tested in accordance with 4.4.6 of HH-I-1751/GEN at the temperature rating specified in the contract or purchase order (see 6.2 and 6.4) thermal insulation sleeving shall show no physical changes that adversely affect its service qualities.

3.1.8 Moisture absorption. When tested in accordance with 4.4.7 of HH-I-1751/GEN thermal insulation sleeving for use below ambient shall gain no more than 1.0 percent by volume.

3.1.9 Corrosiveness. When tested in accordance with 4.4.1 of this specification, the steel test plate in contact with the thermal insulation sleeving shall show no greater corrosion than sterile cotton in contact with a steel plate for comparison.

3.1.10 Linear shrinkage. When tested in accordance with 4.4.2 of this specification at the temperature rating specified in the contract or purchase order (see 6.2 and 6.4) thermal insulation shall show no physical changes that adversely affect its service qualities.

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3.2 Form A, Type 1.

3.2.1 Construction. Molded type pipe and tube covering thermal insulation sleeving shall be made from mineral fibers and binder, molded into sections in the form of hollow cylinders, split lengthwise on one or both sides of the cylindrical axis for application on standard sizes of piping or tubing. Thickness available for various pipe and tube sizes are nominal. Actual thicknesses will vary in accordance with manufacturer's standard sizes. When installed on pipe or tube of specified size outside diameter (see 6.2), sections shall fit snugly and have tight longitudinal and circumferential joints.

3.2.2 Jackets. Unless otherwise specified (see 6.2), a laminated jacket with kraft paper outer surface bonded to a vapor barrier film shall be furnished. The jacket shall have a permeability rating of .02 perms, and a tensile strength of not less than 30 lbs. per inch - width, and shall have a beach puncture resistance of not less than 50 beach units. No jacket shall be funished on the inner layer of double layer construction, except; when said application is for either dual temperature or cold pipe applications.

3.2.3 Bands. Aluminum or steel bands, black jappanned or lacquered, shall be furnished when specified and as specified (see 6.2). All bands shall be at least 1/2-inch wide and not less than 0.005-inch thick. Bands shall be furnished at the rate of five per two sections of pipe covering.

3.3 Form B, Type II.

3.3.1 Construction. Molded pipe and tube fitting covering, thermal insulation sleeving shall be made from mineral fibers and binder, molded to the proper shape and size, and split in segments for application to pipe and tube fittings as specified (see 6.2).

3.3.2 Accessories. Fillers, cement, and other separate or attached accessories for installation shall be as specified (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 The applicable Quality Assurance Provisions of Federal Specification HH-I-1751/GEN and the documents therein, shall apply with the following additional requirements:

4.2 Sampling.

4.2.1 Sampling shall be in accordance with 4.2.1 of HH-I-1751/GEN unless otherwise specified, (see 6.2).

4.3 Examination.

4.3.1 End item. Insulation selected in accordance with 4.2.2 of HH-I-1751/GEN shall be examined for defects listed in table I of HH-I-1751/GEN.

4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking requirements comply with section 5. Defects shall be scored in accordance with table IV of HH-I-1751/GEN.

4.4 Test methods.

4.4.1 Corrosiveness. Test specimen.-Two specimens shall be used, each measuring 1 inch by 4 inches by approximately 1/2 inch thick. Apparatus.- Two polished-steel test plates 1 inch wide, 4 inches long and 0.020 inch thick. They shall be clear finish, cold-rolled strip steel, American quality, quarter hard, temper No. 3, weighing 0.85 lb/sq ft., and humidity test chamber. Procedure.- The steel test plates shall be rinsed with cp benzol until their surfaces are free from oil and grease and allowed to dry. One piece of cold-rolled steel shall be placed between the two insulation specimens and secured with tape or twine. The test specimen and the sterile cotton covered plate shall be suspended veritcally in an atmosphere having a relative humidity of 95 percent ± 3 percent, and a temperature of 120° ± 3°F, for 96 hours, and then be examined for corrosion (see 3.1.9).

4.4.2 Linear shrinkage. Determination of linear shrinkage shall be in accordance with ASTM C 356 (see 3.1.8).

4.4.3 Permeability. Determination of permeability shall be in accordance with ASTM E 96 (see 3.2.2).

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4.4.4 Tensile strength. Determination of tensile strength shall be in accordance with ASTM D 828 (see 3.2.2).

4.4.5 Beach puncture resistance. Determination of beach puncture resistance shall be in accordance with ASTM D 781 (see 3.2.2).

5. PREPARATION FOR DELIVERY. In accordance with Federal Specification HH-I-1751/GEN and documents referenced therein.

6 NOTES

6.1 Intended use. Insulation units covered by this specification are intended, as indicated, for use on surfaces at low and high temperatures.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Form, type, class (see 1.2.1) and temperature rating (see 3.1.7).
(c) Size of insulation, length and thickness (see 1.2.3 and table I).
(d) Form A, Type 1, size, outside diameter of pipe or tube for which molded sleeving is required (see 3.2.1).
(e) Form A, Type I, material and finish for bands when required (see 3.2.3).
(f) Form A, Type I, Type jacket required (see 3.2.2).
(g) Form B, Type II, type and size of pipe or tube fitting for which molded covering is required (see 3.3.1).
(h) Size of lot, if different from 4.2.1, of HH-I-1751/GEN.
(i) Form B, accessories for installation of pipe or tube fittings (see 3.3.2).
(j) Selection of applicable levels of packaging and packing (see 5.1 and 5.2 of HH-I-1751/GEN).

6.3 Temperature rating. Insulation is produced in a large range of temperature ratings. The various classes shown in 1.2.2 indicate a temperature limitation for each specific class. The individual classes cover materials having temperature ratings up to and including the maximum temperature shown; however, these materials do not have to meet this maximum temperature unless specifically required to do so.

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 84

Review activity:

Army - CE
Navy - YD, SH, AS

User activities:

Army - ME
Navy - YD
Air Force - 84

Military Coordinating activity:

Army - ME

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

COMMERCE - NBS
DOT - FIS
GSA - FSS, PCD, TCS
INTERIOR - BPA

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